

Adding New PM Source and Grafana dashboard

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Adding New PM source

Node ID's

Node ID's are used only for Ciena PM collection.

Ciena C6500

Site	Node ID
AUSY-1	81ab0922-f470-38de-a1d5-3a3031d8c2b3
NZMA-1	85ee356a-67a8-3471-8301-a1eb7b24e5d3
HIKA-1	19b4a7c6-253b-33d3-8289-e7202152b6d9
HIKA-2	deb5582c-2835-3257-bf13-8cfaf0f10873
USPA-1	7298a470-0684-367d-ad83-ef346ca9e4e1
USPA-2	ab89ea0e-d6e5-3afb-8657-d22f9d71700b
USFB-1	46147a67-1425-33bf-a847-7c08224a7811

Ciena CWSAI

Site	Node ID
AUSY-1	96f6e6c1-441c-33be-ad2a-3fd0cb21ae9b
AUSY-2	8cb99422-6844-3ac1-b270-0bdc5da1bad4
AUSY-3	6a1144f4-091f-320f-b309-a331095b5239

NZMA-1	7457410e-2692-3274-a1bf-104ef4bc541a
NZMA-2	48c69ad2-ec88-326b-a2e8-1a4e6dfefb829
HIKA-1	09135626-dc41-3c19-ba72-a36216587fd8
HIKA-2	18d37cf6-5102-347d-b35c-8810ed9d6f97
HIKA-3	10eb8bd1-d21e-3f68-8461-3dbf5c9b2adf
USPA-1	61cf31ca-edba-30fd-920a-6316ad31300b
USPA-2	52b12431-4534-3794-9cfa-8e1c1bcc3cdd
USPA-3	d55da130-cbac-3ab1-83ff-15145b73bf55
USFB-1	8232cdea-450b-326e-9ce2-09e5e056e8c2
USFB-2	62a14c72-17aa-3a9c-b4be-23d8404cecbc
USFB-3	9f23802a-43c9-3e72-830b-e44268b0956e

MariaDB Tables

Both Subcom and Ciena PM(6500, CWSAI, CWS5) sources have to be added to MariaDB tables.

service_source

PM sources for TE-Subcom, Ciena 6500 and Ciena-WS has to be added to this table.

Field	Details
id INT(11)	unique id
customer VARCHAR(16)	customer name
customer_sid VARCHAR(16)	customer service id. eg. DEC-BC0074
pm_source VARCHAR(128)	pm source name, eg. Subcom: T5.FP3.HIKA.OCH:192.100:10GbE-3 Ciena: AUSY:PORT-1-7
visible_to_customer TINYINT(4)	flag value set for filtering visibility of PM data in the Grafana dashboard. values: 0 or 1
route_end VARCHAR(3)	the root end is marked with `A` for the south end and `Z` for the north end
segment_end VARCHAR(3)	the segment end is marked in the order `1A`, `2A`... for the south ends and `1Z`, `2Z`... for the north ends in the root
source VARCHAR(16)	the source is the vendor name with the device type, eg: TE-SUBCOM, CIENA, CIENA-WS
node_id VARCHAR(64)	node ids are added only for Ciena services
to_from VARCHAR(2)	this field is not used
start_date DATETIME	start date of the service
end_date DATETIME	the end date of the service
ccode VARCHAR(3)	

new_customer_sid VARCHAR(10)	
---------------------------------	--

customer

Field	Details
id INT(11)	unique id
name VARCHAR(45)	customer name
shortname VARCHAR(45)	service id's first Three alphabets, this short name will be common for all the service id's for a customer
active TINYINT(4)	Flag 0 or 1 for marking active client or not

service

Field	Details
id INT(10)	unique id
serviceid VARCHAR(45)	service id. eg. DEC-BC0074
customer_id INT(11)	foreign key of customer table's id field
startdate DATE	start date of the service
enddate DATE	the end date of the service
bandwidth VARCHAR(45)	bandwidth of this service
active TINYINT(4)	Flag 0 or 1 for marking active service or not
visible_to_customer TINYINT(4)	Flag 0 or 1 for marking this service is visible in the Grafana dashboard or not

configuration

Field	Details
id INT(10)	unique id
serviceid VARCHAR(45)	foreign key of the service table's serviceid field
sourcename VARCHAR(45)	pm source name, eg. Subcom: T5.FP3.HIKA.OCH:192.100:10GbE-3 Ciena: AUSY:PORT-1-7
nodeid VARCHAR(45)	node ids are added only for Ciena services
vendor VARCHAR(45)	the vendor name with the device type, eg: TE-SUBCOM, C6500-1, CWSAI-1
collection_duration INT(11)	collection frequency time interval, normally 15 minutes
route_direction VARCHAR(45)	the root direction is marked with `A` for the south end and `Z` for the north end

segment_direction VARCHAR(45)	the segment direction is marked in the order `1A`, `2A`... for the south ends and `1Z`, `2Z`... for the north ends in the root
nodedisplayname VARCHAR(45)	display name of the node, currently not used
disable_collection TINYINT(4)	Flag 0 or 1 for marking the PM should be collected or not
visible_to_customer TINYINT(4)	Flag 0 or 1 for marking this PM is visible in the Grafana dashboard or not

Steps for adding new PM in DB

Step 1 - Firstly, check whether the service request is for CIENA 6500, CIENA-WS, CIENA-WS5 or TEMS.

For all other requests, fill the following DB tables:

1. Service_source
2. Customer
3. Service
4. Configuration

Step 2 -

For Service_source table:

To check whether the Node-Id provided is correct or not, refer on the confluence page or check through the REST API. After filling the Node-IDs, fill other details such as the route_end and segment information. Each request can have only 1 route beginning and 1 ending.

For Customer table:

Check whether the customer already exists in the customer table, if its a new customer, then add the customer record.

For Service table:

Add one record for each service request. If the serviceid is eg. 'AOZ-BX0013' then the bandwidth would be 10Gb (in roman).

For Configuration table:

Add configuration record for each ports. If the PM_Label is AUSY-C6500-1: SL5 P37 then the sourcename would be AUSY:ETTP-1-5-37.

Steps for creating Grafana dashboard

For new customers:

1. Create a new folder in grafana with the customer name.
2. Create a new dashboard in the created folder and save it. Copy the UID from the JSON code of the dashboard and delete the dashboard.
3. Now, copy the JSON code of any other 8X factor dashboard and replace the customer name with the new customer name.
4. Replace the UID of the previous customer with the new copied UID.
5. Now, import this JSON code in the grafana dashboard and choose the customer name in the folder selection.

Steps for updating Alarm dictionary

Environment details

Vendor	Server	path	Dictionary file
Ciena	10.7.1.41	/opt/HawaikiTools/CienaAlarms	CienaCustomerPMs.csv
TEMS	10.7.1.41	/opt/HawaikiTools/TemsAlarms	TemsCustomerPMs.csv

Steps to follow

- 1, Get port information from the table `service\_source`.

SQL Query for Ciena:

```
select customer, customer_sid, substring(pm_source, 6), node_id FROM pm_db.service_source where source like 'CIENA%' and `visible_to_customer`=1 and `customer_sid` not in (select SUBSTRING_INDEX(customer_sid, ':', -1) from service_source where `customer_sid` like '%::%' and source like 'CIENA%') order by customer, customer_sid
```

The above query gets all the active services that are gathering PM collection in our mediation server from our Maria DB.

We are excluding those services that are sub-services eg., Amazon service "DEC-BC0320" has a Superloop sub-service "EPP-BC0659::DEC-BC0320". In this case, we will not add the Superloop service to the alarm dictionary, as requested by stakeholders, to reduce the amount of alarm emails we get as we are already going to receive the alarm for the primary Amazon service.

Please note: We have now added a new functionality of adding backhaul details from NRMS. After that functionality is deployed to production, we will not need to conduct this activity upon every new addition of a service in our database. Jump to next section for more info.

SQL Query for TEMS: *SELECT customer, customer_sid, pm_source FROM pm_db.service_source where source = 'TE-SUBCOM' order by customer, customer_sid;*

- 2, Save the data into a file as `;` separated CSV

- 3, Remove the top row(field names).

Ensure that the file is in Unix (LF) Format.

- 4, Open dictionary file from the server.

- 5, Update the dictionary file with the new rows.

Alarms Sending Algorithm

- Alarms sending logical algorithm for Ciena Devices

We only gather alarms with severity critical and major.

If these alarms belong to our Service IDs (by checking in Customer PMs file) and are critical, we email them.

We send all critical and major alarms to OpenNMS.

- Alarms sending logical algorithm for TEMS Subcom Devices

We gather all alarms.

If these alarms belong to our Service IDs (by checking in Customer PMs file), we email them.

We send all alarms to OpenNMS.

Backhaul Information Addition to Alarm Emails

We have created a job that will gather the Backhaul and Service Id data from NRMS and our own mediation database on a regular basis.

The jobs are located at : "/opt/HawaikiTools/CienaAlarms/ciena-customerpms-dictionary" on the server : 10.7.1.41

The shell script needs to run to gather backhaul data json from NRMS, which would be stored in a Json file "file-new.json".

```

3  {
    "customername": "AVELACOM",
    "vendor": "CIENA",
    "shelf": "1",
    "slot": "7",
    "dis": "DLS14",
    "portname": "AUSY/FOTR/010201.04/S_1:Sl_7:P_18-Rx:C_LCU",
    "servicename": "NMO-BX0470",
    "route": "AU-HI-001",
    "segment": "AU-HI-14",
    "offnetvendor": "Hawaiian Tel",
    "offnetid": "88.PIGS.100030..HAWT",
    "offnetprotected": "false",
    "offnetaddetails": "HIKA: RefId:82.PIGS.900980..HAWT, RR: CAB04 , LLM:29 , PORT: A3,A4",
    "offnetbenddetails": "USLA: RefId:MKC/HTC-LAP/HTC 10G 10G00208, RR: ODF01\\t, Panel: 3 , Port: 122",
    "portnumber": "18"
  },
3  {
    "customername": "AVELACOM",
    "vendor": "CIENA",
    "shelf": "1",
    "slot": "7",
    "dis": "DLS14",
    "portname": "AUSY/FOTR/010201.04/S_1:Sl_7:P_18-Tx:C_LCU",
    "servicename": "NMO-BX0470",
    "route": "AU-HI-001",
    "segment": "AU-HI-14",
    "offnetvendor": "Hawaiian Tel",
    "offnetid": "88.PIGS.100030..HAWT",
    "offnetprotected": "false",
    "offnetaddetails": "HIKA: RefId:82.PIGS.900980..HAWT, RR: CAB04 , LLM:29 , PORT: A3,A4",
    "offnetbenddetails": "USLA: RefId:MKC/HTC-LAP/HTC 10G 10G00208, RR: ODF01\\t, Panel: 3 , Port: 122",
    "portnumber": "18"
  },
}

```

The shell script titled "ciena_pm.sh" will then run the Pentaho job that will gather data from our mediation database as the above data from NRMS does not contain all the details. It will then parse the above Json, and correlate both the files' data to create the **finalized PM dictionary** file with the name "CienaCustomerPMs.csv".

```

94  AMAZON:DEC-BC0920:::AMAZON:PORT-3-7:61cf31ca-edba-30fd-920a-6316ad31300b
95  AMAZON:DEC-BC0920:::AMAZON:PORT-3-13:5ff56004-538d-3233-9482-70fb10b99383
96  AMAZON:DEC-BC0920:::AMAZON:PORT-2-3:57ca4411-2e4e-38a7-aa9d-dd134021f27b
97  AMAZON:DEC-BC0920:::AMAZON:PORT-3-13:424dfe6-3c80-390b-a920-91ce1225f5e9
98  AMAZON:DEC-BC0924:::AMAZON:PORT-3-14:ace22ea4-35c3-391c-b05a-65935d4a7d46
99  AMAZON:DEC-BC0924:::AMAZON:PORT-3-14:7d51e007-f525-3d9a-82d7-85f13990f719
100 AMAZON:DEC-BC0924:::AMAZON:PORT-3-14:5ff56004-538d-3233-9482-70fb10b99383
101 AMAZON:DEC-BC0924:::AMAZON:PORT-3-14:424dfe6-3c80-390b-a920-91ce1225f5e9
102 SALESFORCE:EEE-BX0491:88.PIGS.100045..HAWT:Hawaiian Tel:false;HIKA: RefId:82.PIGS.900981..HAWT, RR: CAB04 , LLM:29 , PORT: A9,A10;USLA: RefId:MKC/HTC-LAP/HTC 10G 10G00301, RR: ODF01 , Panel: 3 , Port: 129;SALESFORCE:ETTP-1-7-38;81ab0922-f470-38de-aid5-3a3031d8c2b3
103 SALESFORCE:EEE-BX0491:88.PIGS.100045..HAWT:Hawaiian Tel:false;HIKA: RefId:82.PIGS.900981..HAWT, RR: CAB04 , LLM:29 , PORT: A9,A10;USLA: RefId:MKC/HTC-LAP/HTC 10G 10G00301, RR: ODF01 , Panel: 3 , Port: 129;SALESFORCE:ETTP-1-7-38;19b4a7c6-253b-33d3-8289-e7202152b6d9

```

The newly created Customer PMs dictionary file

The Customer PMs dictionary created above, is different from our current dictionary file i.e the new file consists of the newly added backhaul data. As seen above, only some services have additional backhaul information available in our network.

Things to be added :

- Modify the current shell script file titled "CienaAlarms.sh" at "/opt/HawaikiTools/CienaAlarms" to add the newly created backhaul details columns in alert emails.

